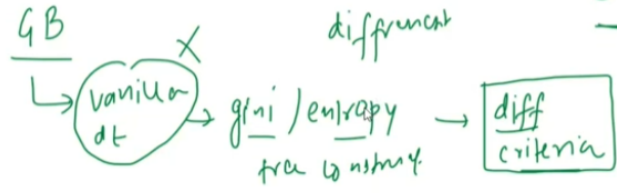
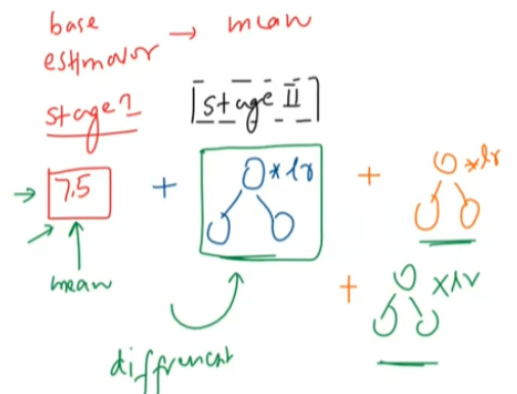
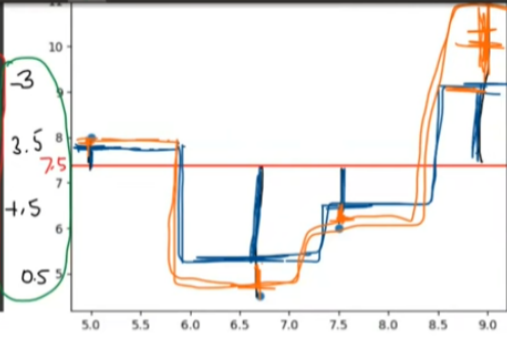


cgpa	package
6.7	4.5
9.0	11.0
7.5	6.0
5.0	8.0



Step by Step Calculation

23 October 2023 15:07

7.3

stage 1 → mean

cgpa	package	model 1	residual 1
6.7	4.5	7.3	-2.8
9.0	11.0	7.3	3.7
7.5	6.0	7.3	-1.3
5.0	8.0	7.3	0.7



cgpa | residual 1

decision tree

↓ ↓ ↓ ↓

-2.8, 3.7, -1.3, 0.7 → Similarity score

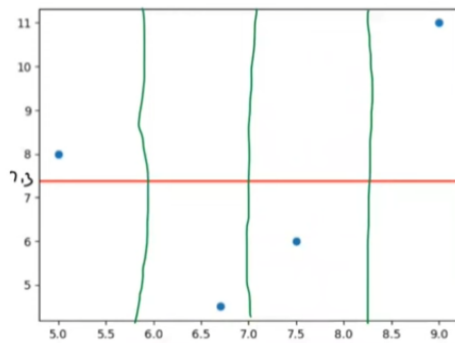
$$SS = \frac{(\text{sum of residuals})^2}{\# \text{residuals} + 1}$$

λ = 0

→ reg parameter

$$\frac{(-2.8 + 3.7 - 1.3 + 0.7)^2}{4 + 0} = \frac{(0.3)^2}{4} = 0.02$$

6.7	4.5	7.3	-2.8
9.0	11.0	7.3	3.7
7.5	6.0	7.3	-1.3
5.0	8.0	7.3	0.7



$$SS = \frac{(\text{sum of residuals})^2}{\# \text{residuals} + 1}$$

$$\frac{(-2.8 + 3.7 - 1.3 + 0.7)^2}{4 + 0} = \frac{(0.3)^2}{4} = 0.02$$

$\lambda = 0$

Similarity score

$$\begin{aligned} 5.0 &\rightarrow 5.85 \\ 6.7 &\rightarrow 7.1 \\ 7.5 &\rightarrow 8.25 \\ 9.0 &\rightarrow \end{aligned}$$

Splitting criteria ① $\rightarrow$ 5.85

$\rightarrow$ $\boxed{cgpn < 5.85}$

$\boxed{0.7}$

$$\begin{aligned} \underline{SS_L} &= \frac{(0.7)^2}{1+0} \\ &= \underline{0.49} \end{aligned}$$

$\boxed{-2.8, -1.3, 3.7}$

$$\begin{aligned} \underline{SS_R} &= \frac{(-2.8 - 1.3 + 3.7)^2}{3} \\ &= \frac{0.16}{3} = \underline{0.05} \end{aligned}$$

$$\text{gain} = (SS_L + SS_R) - SS_{\text{root}}$$

Splitting criteria ② $\rightarrow$ 7.1

Splitting criteria ③ $\rightarrow$ 0.1

Splitting criteria ① $\rightarrow 5.85$

$\rightarrow \boxed{cgpa < 5.85} \checkmark$

$$\begin{array}{l}
 \boxed{0.7} \\
 \downarrow \\
 SS_L = \frac{(0.7)^2}{1+0} \\
 = 0.49
 \end{array}
 \quad
 \begin{array}{l}
 \boxed{-2.8, -1.3, 3.7} \\
 \downarrow \\
 SS_R = \frac{(-2.8 - 1.3 + 3.7)^2}{3} \\
 = \frac{0.16}{3} = 0.05
 \end{array}$$

$$\begin{aligned}
 \text{gain} &= (SS_L + SS_R) - SS_{\text{root}} \\
 &= 0.49 + 0.05 - 0.02 \\
 &= \boxed{0.52} \checkmark
 \end{aligned}$$

Splitting criteria ② $\rightarrow 1.1$

$\rightarrow \boxed{cgpa < 7.2} \rightarrow \text{Similar}$

$$\begin{array}{l}
 \boxed{0.7, -2.8} \quad \boxed{-1.3, 3.7} \\
 \downarrow \quad \quad \downarrow \\
 SS_L = \frac{(0.7 - 2.8)^2}{2} \quad SS_R = \frac{(-1.3 + 3.7)^2}{2} \\
 = \frac{4.41}{2} \quad = \frac{5.76}{2} \\
 = 2.20 \quad = 2.88
 \end{array}$$

$$\begin{aligned}
 \text{gain} &= SS_L + SS_R - SS_{\text{root}} \\
 &= 2.20 + 2.88 - 0.02 \\
 &= \boxed{5.06} \checkmark
 \end{aligned}$$



Splitting criteria ① $\rightarrow 5.85$

$\rightarrow \boxed{\text{gpa} < 5.85} \checkmark$

$\boxed{0.7}$

$$SS_L = \frac{(0.7)^2}{1+0} = 0.49$$

$\boxed{-2.8, -1.3, 3.7}$

$$SS_R = \frac{(-2.8 - 1.3 + 3.7)^2}{3} = \frac{0.16}{3} = 0.05$$

$$\begin{aligned} \text{gain} &= (SS_L + SS_R) - SS_{\text{root}} \\ &= 0.49 + 0.05 - 0.02 \\ &= \boxed{0.52} \checkmark \end{aligned}$$

Splitting criteria ② $\rightarrow 7.1$

$\rightarrow \boxed{\text{gpa} < 7.1} \rightarrow \text{Similar}$

$\boxed{0.7, -2.8}$

$$\begin{aligned} SS_L &= \frac{(0.7 - 2.8)^2}{2} \\ &= \frac{4.41}{2} \\ &= 2.20 \end{aligned}$$

$\boxed{-1.3, 3.7}$

$$\begin{aligned} SS_R &= \frac{(-1.3 + 3.7)^2}{2} \\ &= \frac{5.76}{2} \\ &= 2.88 \end{aligned}$$

$$\begin{aligned} \text{gain} &= SS_L + SS_R - SS_{\text{root}} \\ &= 2.20 + 2.88 - 0.02 \\ &= \boxed{5.06} \checkmark \end{aligned}$$

Splitting criteria ③ $\rightarrow 8.25$

$\boxed{\text{gpa} < 8.25} \checkmark$

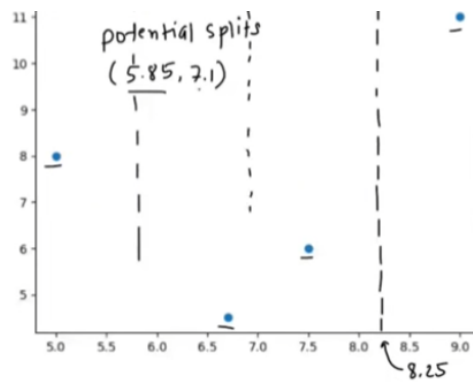
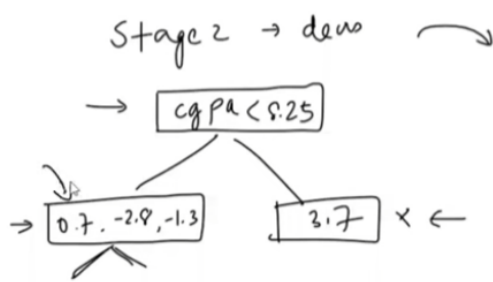
$\boxed{0.7, -2.8, -1.3}$

$$\begin{aligned} SS_L &= \frac{(0.7 - 2.8 - 1.3)^2}{3} \\ &= \frac{11.56}{3} \\ &= 3.85 \end{aligned}$$

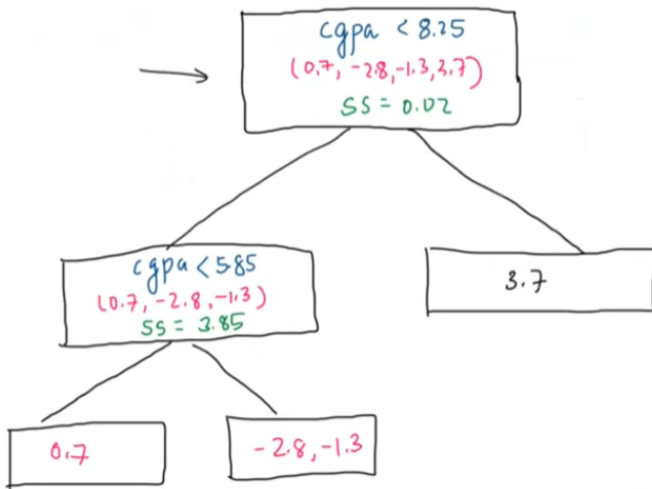
$\boxed{3.7}$

$$\begin{aligned} SS_R &= \frac{(3.7)^2}{1} \\ &= 13.69 \end{aligned}$$

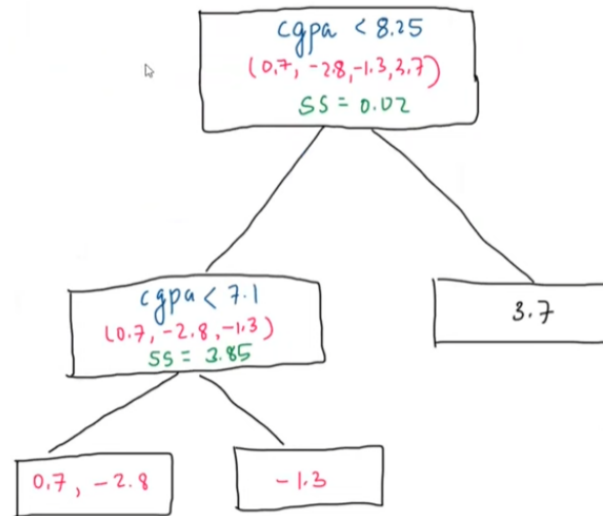
$$\begin{aligned} \text{gain} &= 3.85 + 13.69 - 0.02 \\ &= \boxed{17.52} \end{aligned}$$

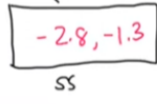


① Splitting Criteria $\rightarrow 5.85$



② Splitting criteria $\rightarrow 7.1$

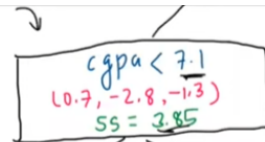




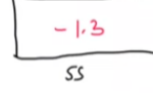
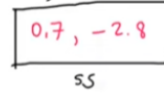
$$SS_L = \frac{(0.7)^2}{1} = 0.49$$

$$SS_R = \frac{(-2.8 - 1.3)^2}{2} = \frac{(-4.1)^2}{2} = 8.40$$

$$\text{gain} = (SS_L + SS_R - SS_P) = 0.49 + 8.40 - 3.85 = 5.04$$



3.7

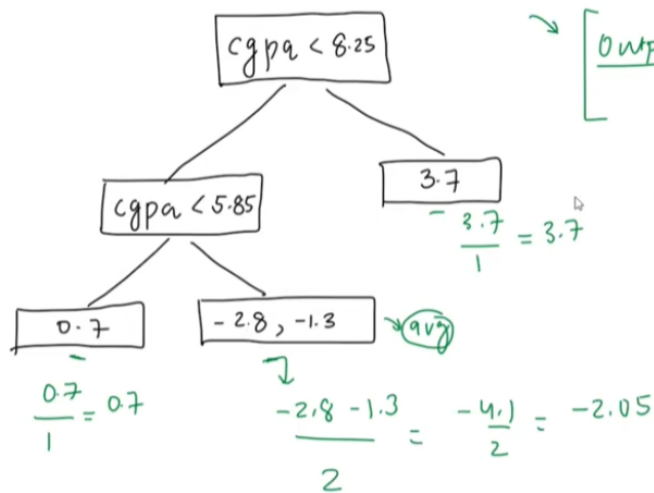


$$SS_L = \frac{(0.7 - 2.8)^2}{2} = \frac{(-2.1)^2}{2} = 2.20$$

$$SS_R = \frac{(-1.3)^2}{1} = 1.69$$

$$= 2.20$$

$$\text{gain} = 2.20 + 1.69 - 3.85 = 0.04$$



$$\rightarrow \left[\text{Output} = \frac{\text{sum of residuals}}{\# \text{residuals} + \lambda} \right]$$

$$\lambda = 0 \rightarrow \lambda = 0 \rightarrow \text{avg}$$

	cgpa	package	model 1	residual 1	model 2	residual 2
→	6.7	4.5	7.3	-2.8	6.69	-2.19
→	9.0	11.0	7.3	3.7	8.41	2.59
→	7.5	6.0	7.3	-1.3	6.69	-0.69
→	5.0	8.0	7.3	0.7	7.51	0.49

Stage (3)
 $\text{cgpa} / \text{res 2}$
 $\text{mean} + \text{lr} \times \text{dt1} + \text{lr} \times \text{dt2}$

$$7.3 + 0.3 \times (-2.05)$$

$$7.3 + 0.3 \times 3.7$$

$$7.3 + 0.3 \times (-2.05)$$

$$7.3 + 0.3 \times 0.7$$

